

AB RENEWABLE ENERGY ENGINEERING

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BIOMASS GASIFICATION

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1. Which of the following is not a biomass?

- a. Corn cobs
- b. Rice husks
- c. Peat
- ☒ d. None of the above

2. Product resulting from subjecting biomass to thermo-chemical reaction.

- ☒ a. Producer gas
- b. Biodiesel
- c. Bioethanol
- d. Biogas
- e. None of the above

3. If high percentage of carbon is required from biomass, which of the following thermo-chemical process would you recommend?

- a. Direct combustion
- ☒ b. Pyrolysis
- c. Gasification
- d. All of the above

4. Heating value of producer gas is typically at ____.

- ☒ a. 1200 kcal/m³
- b. 2,245 kcal/m³
- c. 3,000 kcal/m³
- d. None of the above

5. Type of engine that can be fueled with producer gas.

- a. Gas engine
- b. Diesel engine
- c. Gasoline engine
- ☒ d. All of the above

6. Biomass-gas engine is basically a modified ____ engine.

- a. gasoline
- ☒ b. diesel
- c. steam
- d. None of the above

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7. Percentage amount of diesel fuel that can be replaced with gas producer.

- a. 20-40 %
- ☒ b. 50-70 %
- c. 90-100 %
- d. None of the above

8. At an arbitrary speed, the power output of a diesel engine is reduced by ____ when it is powered with gas producer.

- ☒ a. 10-20 %
- b. 30-50 %
- c. 50-70 %
- d. None of the above

9. Gases produced from producer gas.

- ☒ a. CO₂ and CH₄
- b. CO, H₂, and CH₄
- c. CO₂
- d. All of the above
- e. None of the above

10. Amount of air needed in pyrolysis biomass.

- ☒ a. Below 30% of stoichiometric air
- b. 30 to 40 % of stoichiometric air
- c. Above 40% of the stoichiometric air
- d. None of the above

11. Equivalence ratio for pyrolyzers.

- ☒ a. Up to - 0.2
- b. 0.2 - 0.4
- c. 0.4 and above
- d. None of the above

12. Equivalence ratio for biomass gasifiers.

- a. 0 - 0.2
- ☒ b. 0.2 - 0.4
- c. 0.4 and above
- d. None of the above

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13. What is the required airflow to gasify rice husk in a gasifier that consumes 40 kg per hour of fuel? The stoichiometric air requirement of rice husks is 4.7 kgair/kgfuel. Consider an equivalence ratio of 0.3.

- a. 56 kg of air per hr
- b. 70 kg of air per hr
- c. 86 kg of air per hour
- d. None of the above

14. Channel formation at the fuel bed inside the gasifier reactor basically is the result of ____.

- a. high superficial velocity
- b. overloading of fuel during operation
- c. reactor operating at a low temperature
- d. None of the above

15. To minimize channel formation inside the gasifier reactor, the superficial gas velocity in the char bed should not exceed ____.

- a. 8-9 cm/sec

- b. 15-20 cm/sec
- c. 20-23 cm/sec
- d. None of the above

16. Conversion process of solid biomass to combustible gaseous fuel through thermochemical reaction.

- a. Digestion
- b. Carbonization
- c. Gasification
- d. None of the above

17. Primary gas produced during gasification.

- a. Methane
- b. Carbon dioxide
- c. Carbon monoxide
- d. All of the above

18. Amount of air needed in gasifying biomass.

- a. Below 30% of stoichiometric air
- b. 30 to 40 % of stoichiometric air
- c. Above 40% of the stoichiometric air
- d. None of the above

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19. A corn cob has an stoichiometric air requirement of 5.7 kg air/kg fuel. If 30 kg of corn cobs is required to be burned per hour, how much air in m³/hr is required for the system. Consider an equivalence ratio of 0.25

- a. 30.68 m³/hr
- b. 31.28 m³/hr
- c. 34.2 m³/hr
- d. None of the above

20. Gasifier zone where gasification takes place.

- a. Drying zone
- b. Distillation zone
- c. Reduction zone
- d. None of the above

21. Gasification rate of rice hull ranges from ____.

- a. 90-105 kg/m²-hr
- b. 110-210 kg/m²-hr
- c. 125-140 kg/m²-hr
- d. None of the above

22. A rice husk gasifier is operating at 20kg-per-hour fuel rate. What is the diameter required for the gasifier if the designed specific gasification rate is 160 kg/hr-m²?

- a. 0.20 m
- b. 0.25 m
- c. 0.39 m
- d. None of the above

23. Type of gasifiers suitable for rice husk gasifier operating on a continuous mode.

- a. Fixed bed
- b. Moving-bed
- c. Fluidized bed
- d. None of the above

24. Engine suitable for fueling biomass producer gas.

- a. Gasoline engine
- b. Diesel engine
- c. Gas engine
- d. All of the above

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